SAM 3 crop-mark detection

Launcher: `Toolbox/sam3_detection.bat`

This optional tool requires the separate `env_sam3` environment, checkpoint access, and SAM 3 runtime described in README file.

Purpose

SAM 3 applies text-guided segmentation to an RGB rendering of either one GeoTIFF or every GeoTIFF in a folder. It returns candidate masks matching a phrase such as `crop mark`. The result is a model proposal, not a classification of archaeological origin.

Parameters

Option	Meaning and behaviour
Input GeoTIFF or folder	Select one `.tif`/`.tiff` image, or select a folder to scan it non-recursively for all `.tif` and `.tiff` files, case-insensitively. Existing SAM output rasters and partial files are excluded from folder scans.
Output folder	Blank creates `sam3_crop_marks` inside the selected folder. For a single image, it is created beside the image. A specified folder is created if necessary.
Text prompt	Description sent to SAM 3. Default: `crop mark`. It cannot be empty. Try concrete visible concepts, but record the prompt because it materially affects the output.
Minimum confidence	Accepts scores greater than 0 and at most 1. Default: `0.70`. Raising it usually reduces detections and false positives; lowering it increases both.

Option	Meaning and behaviour
RGB bands (1-based)	Exactly three comma-separated positive band numbers in red, green, blue order. Default: `1,2,3`. For WV3 ordered C,B,G,Y,R,RE,N,N2, natural colour is `5,3,2`.
Tile size	Square processing size in pixels. Minimum 256; default `1008`. Larger tiles provide more context but use more GPU memory. Edge tiles are included.
Overlap	Shared pixels between neighbouring tiles. Default `128`; must be non-negative and smaller than tile size. More overlap reduces boundary misses but increases time and duplicate CSV candidates.
Local checkpoint	Optional `.pt` or `.pth`. Blank loads the gated `facebook/sam3` checkpoint through Hugging Face authentication.
Overwrite existing outputs	Off by default. Without it, an image is skipped only when all three final outputs already exist. With it, outputs are replaced after successful processing.

Each RGB tile is independently stretched between its 2nd and 98th valid-data percentiles. This improves display contrast but means the model does not receive calibrated reflectance values and adjacent tiles may have different stretches.

Outputs per image

- `<name>_crop_marks.tif` — UInt8 union mask: 1 where any accepted instance occurs, 0 elsewhere/NoData.
- `<name>_crop_mark_confidence.tif` — Float32 maximum accepted score at each detected pixel.
- `<name>_crop_mark_instances.csv` — one row per tile-level instance with score, tile origin, and pixel bounding box.

Overlapping tiles can report the same feature more than once in the CSV. The raster products merge overlaps by maximum/union. Raster metadata records the prompt, threshold, and RGB bands.

Interpretation and tuning

Begin with the default threshold and overlap on a representative image, then inspect false positives and missed features before processing a large collection. Use the confidence raster for prioritization, not proof. Compare candidates with field boundaries, vegetation patterns, terrain, source resolution, and independent archaeological evidence. CPU inference is supported but very slow.